

Session – 7 : Functional Programming using filter() –

7a- Remove Element -:

```
class Solution {  
    public int removeElement(int[] nums, int val) {  
        int k = 0;  
        for (int i = 0; i < nums.length; i++) {  
            if (nums[i] != val) {  
                nums[k] = nums[i];  
                k++;  
            }  
        }  
        return k;  
    }  
}
```

Output :

The screenshot displays the LeetCode submission page for the 'Remove Element' problem. The interface is in dark mode. At the top, there's a navigation bar with 'Problem List', 'Accepted', 'Editorial', 'Submissions', and 'Solutions'. The 'Accepted' tab is selected, showing '116 / 116 testcases passed' and the submission by 'Praveen_1437' on Sep 15, 2026 11:10. A green 'Solution' button is visible. Below this, a promotional banner for 'Unlock the Full LeetCode Experience' is shown. The main section features a 'Runtime' and 'Memory' performance summary: '0 ms | Beats 100.00%' and '43.34 MB | Beats 78.73%'. A bar chart shows the runtime performance relative to other submissions. The 'Code' section on the right displays the Java code for the 'removeElement' method. The 'Testcase' section on the right shows the 'Test Result' for 'Case 1' and 'Case 2', both of which are 'Accepted'. The input for Case 1 is 'nums = [3,2,2,3]' and 'val = 3', with the output being '[2,2]'. The expected output is also '[2,2]'.

```
class Solution {  
    public int removeElement(int[] nums, int val) {  
        int k = 0;  
        for (int i = 0; i < nums.length; i++) {  
            if (nums[i] != val) {  
                nums[k] = nums[i];  
                k++;  
            }  
        }  
        return k;  
    }  
}
```

Accepted Runtime: 0 ms

Case 1 Case 2

Input

nums =
[3,2,2,3]

val =
3

Output

[2,2]

Expected

[2,2]

7b-Remove Duplicates from Sorted Array :

Program :

```
class Solution {  
    public int removeDuplicates(int[] nums) {  
        if (nums.length == 0) return 0;  
        int k = 1; // first element is always unique  
        for (int i = 1; i < nums.length; i++) {  
            if (nums[i] != nums[k - 1]) {  
                nums[k] = nums[i];  
                k++;  
            }  
        }  
        return k;  
    }  
}
```

Output :

The screenshot shows a LeetCode submission interface. On the left, the 'Accepted' status is confirmed for a submission by 'Praveen_1437' at 11:07 on Sep 15, 2026. A promotional banner for LeetCode's full experience is visible. Below this, a performance graph shows the runtime (0 ms) and memory (0.00 MB) usage, both of which are optimal. A bar chart compares the submission's performance to other users, showing it is faster than most. The code editor on the right displays the Java solution, which is identical to the one provided in the text. The test results section shows that the solution passed all test cases, including Case 1 and Case 2, with an input of [1,1,2] and an expected output of [1,2].

Accepted Praveen_1437 submitted at Sep 15, 2026 11:07

Analysis Solution

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Runtime: 0 ms | Beats: -%
Memory: 0.00 MB | Beats: -%

Code | Java

```
1 class Solution {  
2     public int removeDuplicates(int[] nums) {  
3         if (nums.length == 0) return 0;  
4         int k = 1; // first element is always unique  
5         for (int i = 1; i < nums.length; i++) {  
6             if (nums[i] != nums[k - 1]) {  
7                 nums[k] = nums[i];  
8                 k++;  
9             }  
10        }  
11        return k;  
12    }  
13 }
```

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 Case 2

Input
nums =
[1,1,2]

Output
[1,2]

Expected
[1,2]